

Key Factors Influencing the Price of Ford Vehicles

Final Project Prediction Model

AUTHOR

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	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize
1	Fiesta	2017	12000	Automatic	15944	Petrol	150	57.7	1.0
2	Focus	2018	14000	Manual	9083	Petrol	150	57.7	1.0
3	Focus	2017	13000	Manual	12456	Petrol	150	57.7	1.0
4	Fiesta	2019	17500	Manual	10460	Petrol	145	40.3	1.5
5	Fiesta	2019	16500	Automatic	1482	Petrol	145	48.7	1.0
6	Fiesta	2015	10500	Manual	35432	Petrol	145	47.9	1.6
7	Puma	2019	22500	Manual	2029	Petrol	145	50.4	1.0
8	Fiesta	2017	9000	Manual	13054	Petrol	145	54.3	1.2
9	Kuga	2019	25500	Automatic	6894	Diesel	145	42.2	2.0
10	Focus	2018	10000	Manual	48141	Petrol	145	61.4	1.0

BUSINESS UNDERSTANDING

Business Problem: Jordan Felich, a sales agent at the AutoTrade S.A., leading automotive retailer specializing in used cars, is striving to enhance the company's pricing strategy. With the goal of accurately purchasing and selling vehicles at the most competitive and precise prices, Jordan is focusing on optimizing the pricing process for Ford vehicles. Jordan manages Ford's inventory, which includes vehicles from 1995 to 2020. Jordan intends to use data on critical characteristics such as fuel type, year of production, transmission type, engine size, MPG, and tax to improve pricing accuracy using data-driven insights and predictive models. This effort aims to increase profitability while also ensuring customer happiness by matching prices with market trends and vehicle value.

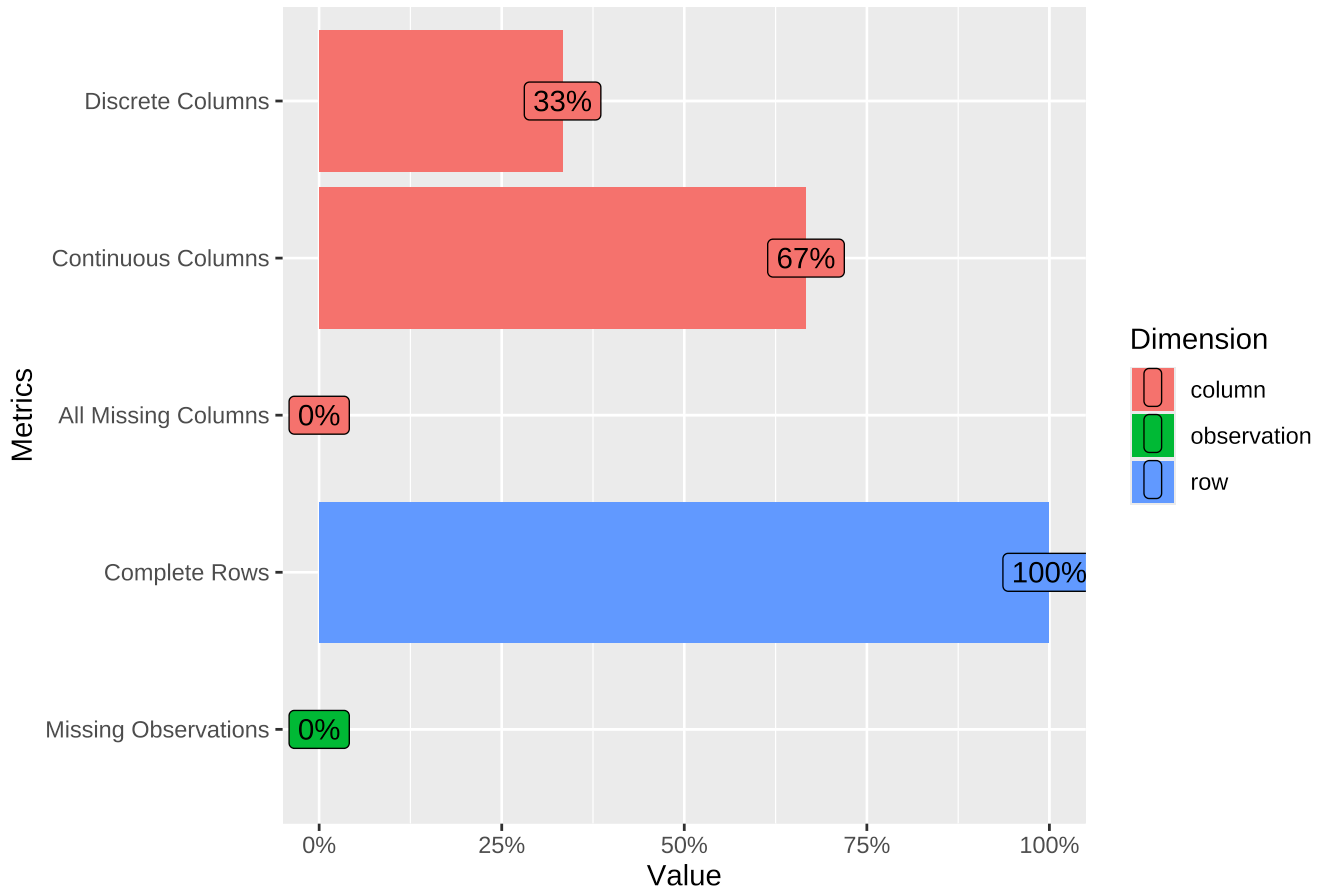
Research Questions:

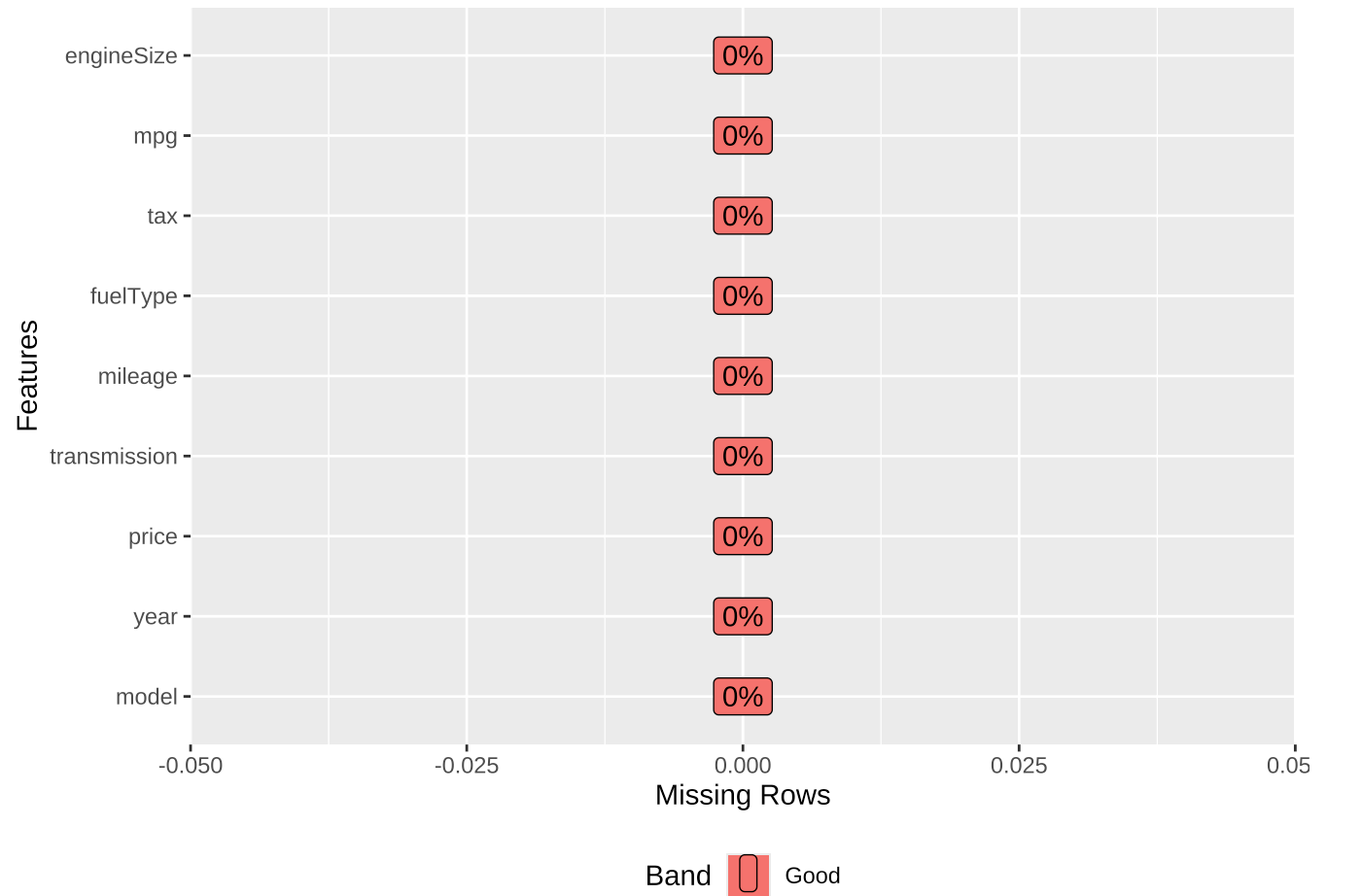
1- What are the key factors influencing the price of Ford vehicles? 2- How can AutoTrade S.A, improve their pricing and marketing strategies, with the information provided by the prediction models?

DATA UNDERSTANDING

EDA Inspect dataset for Missing Values

Memory Usage: 987.2 Kb



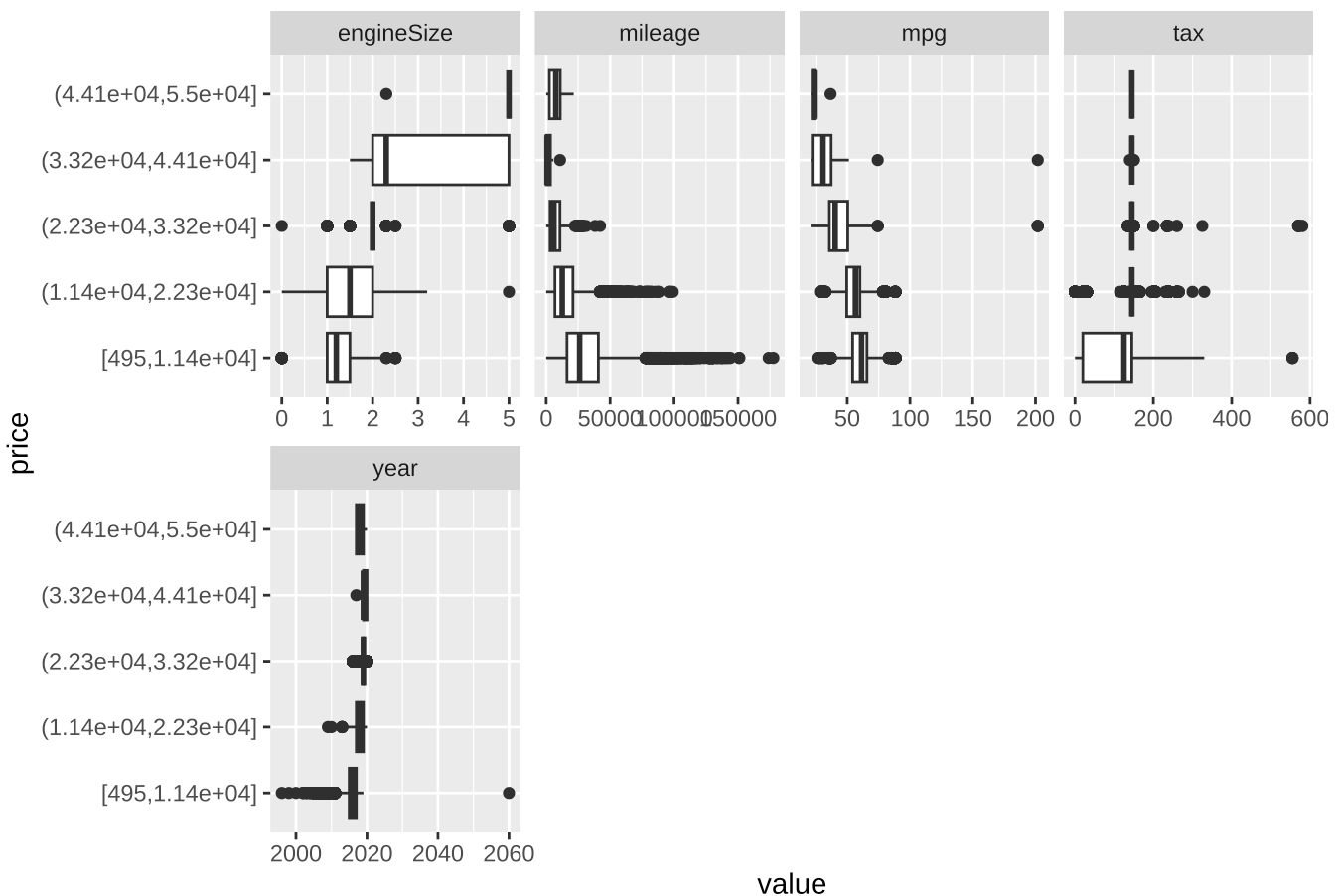


variables	outliers_cnt	outliers_ratio	outliers_mean	with_mean	without_mean
year	499	2.7774686	2009.671343	2016.866470	2017.072021
price	286	1.5918958	29341.685315	12279.534844	12003.529468
mileage	878	4.8870088	81887.873576	23362.608761	20355.517088
tax	20	0.1113214	470.500000	113.329456	112.931405
mpg	216	1.2022710	55.129167	57.906980	57.940783
engineSize	190	1.0575532	2.340526	1.350807	1.340228

EDA Inspect dataset for Outliers

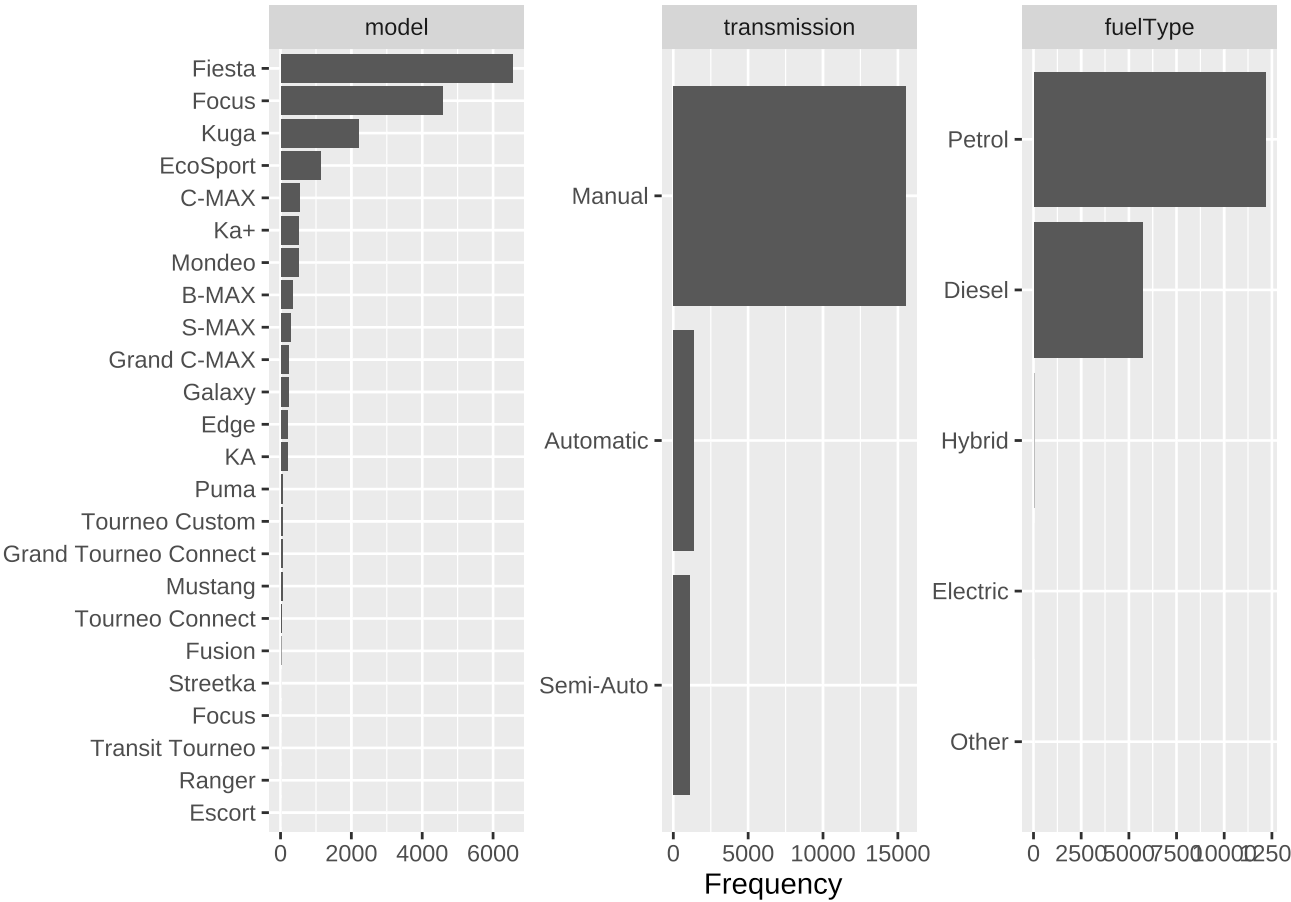
variables	outliers_cnt	outliers_ratio	outliers_mean	with_mean	without_mean
year	499	2.7774686	2009.671343	2016.866470	2017.072021
price	286	1.5918958	29341.685315	12279.534844	12003.529468
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	without_mean				
1	2017.072022				
2	12003.529468				
3	20355.517088				
4	112.931405				
5	57.940783				
6	1.340228				



There is no **missing values** in the data set, this implies that the data is clean in this aspect and there is no needed process needed to be performed. On the other hand, **mileage** and **year**, exhibit the highest number of outliers, however, this could be expected as we are working with a wide range of manufacturing years (1995-2020) and the natural variability in vehicle mileage due to factors such as age and use patterns. At this stage, there is no immediate steps needed, but this outliers will be closely monitored to ensure that they do not affect the results of the study.

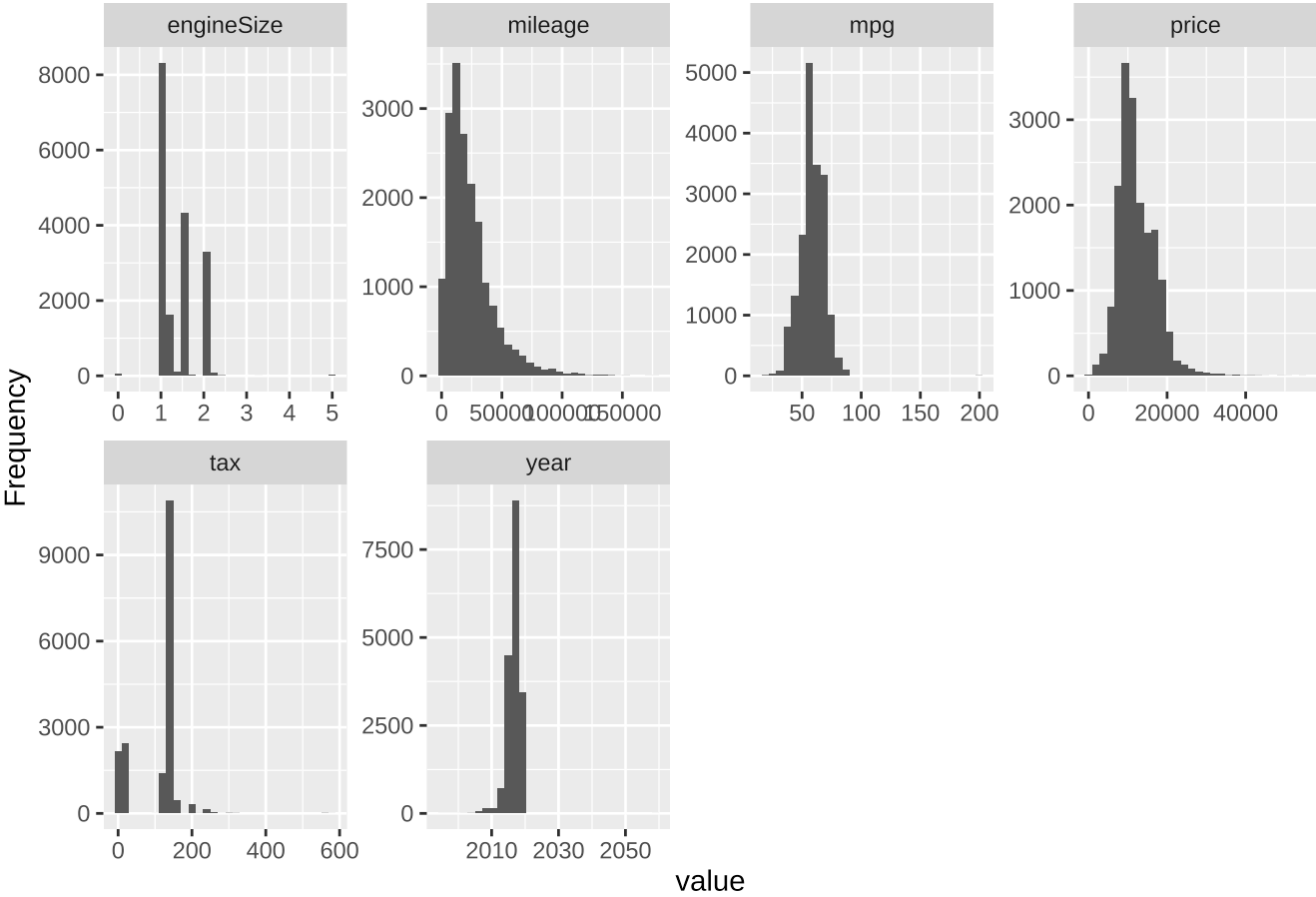
EDA: Look at variable relationships

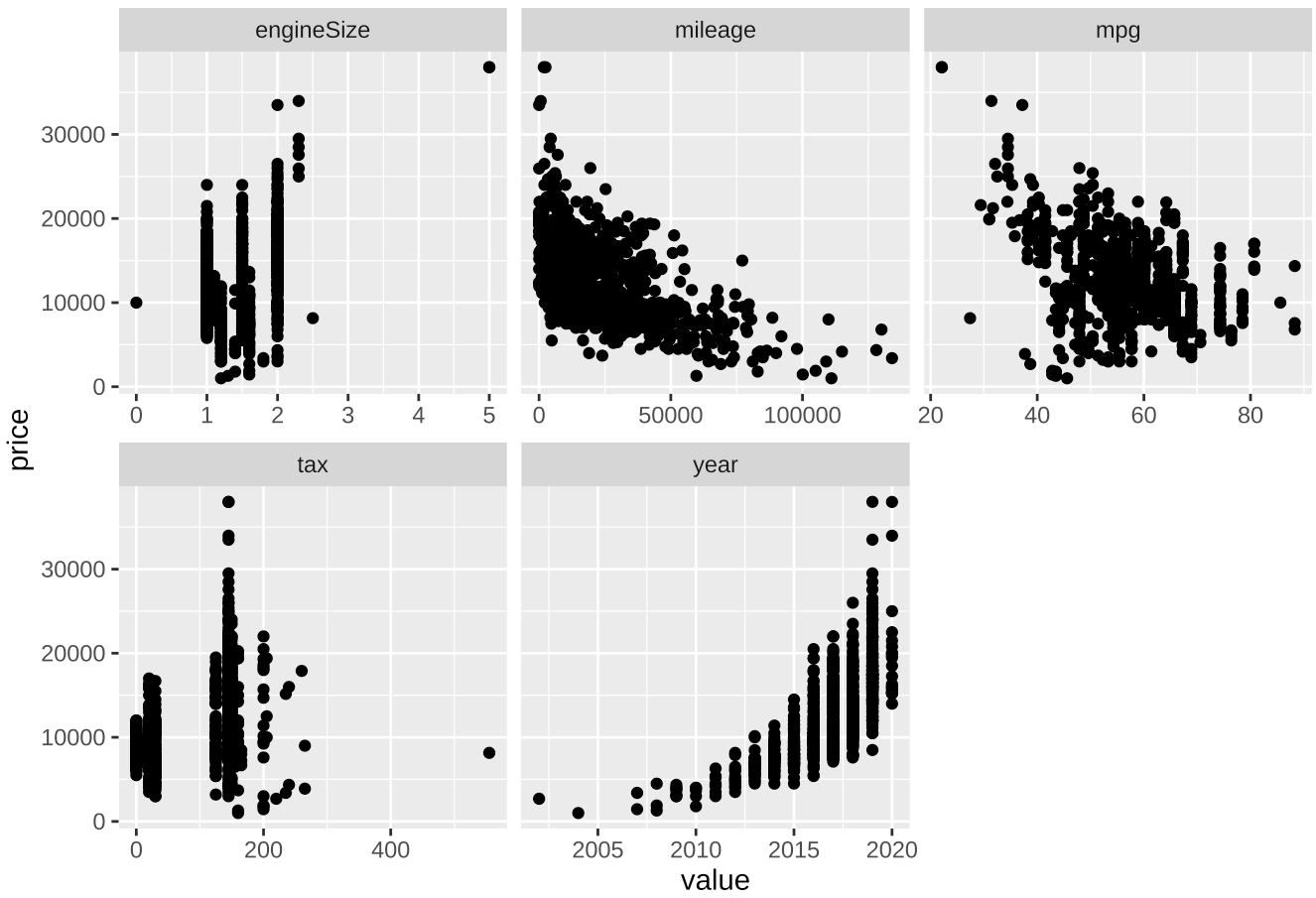


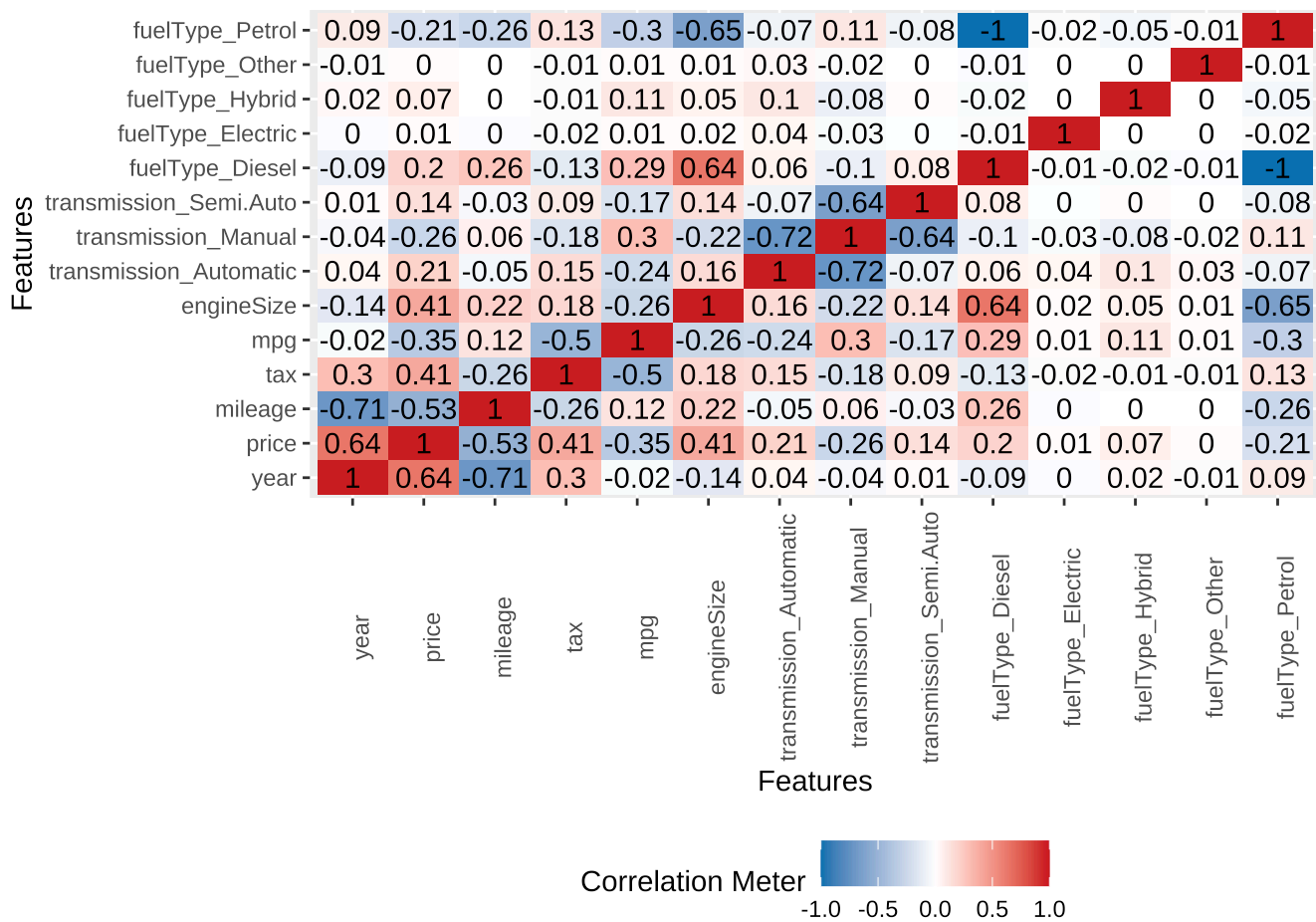
Descriptive Statistics

mydata
N: 17966

	engineSize	mileage	mpg	price	tax	year
Mean	1.35	23362.61	57.91	12279.53	113.33	2016.87
Std.Dev	0.43	19472.05	10.13	4741.34	62.01	2.05
Min	0.00	1.00	20.80	495.00	0.00	1996.00
Q1	1.00	9987.00	52.30	8999.00	30.00	2016.00
Median	1.20	18242.50	58.90	11291.00	145.00	2017.00
Q3	1.50	31064.00	65.70	15299.00	145.00	2018.00
Max	5.00	177644.00	201.80	54995.00	580.00	2060.00
MAD	0.30	14613.25	9.79	4227.63	7.41	1.48
IQR	0.50	21073.00	13.40	6300.00	115.00	2.00
CV	0.32	0.83	0.17	0.39	0.55	0.00
Skewness	1.89	1.84	0.60	1.10	-0.56	-1.28
SE.Skewness	0.02	0.02	0.02	0.02	0.02	0.02
Kurtosis	11.29	4.99	11.43	3.26	1.31	17.03
N.Valid	17966.00	17966.00	17966.00	17966.00	17966.00	17966.00
Pct.Valid	100.00	100.00	100.00	100.00	100.00	100.00







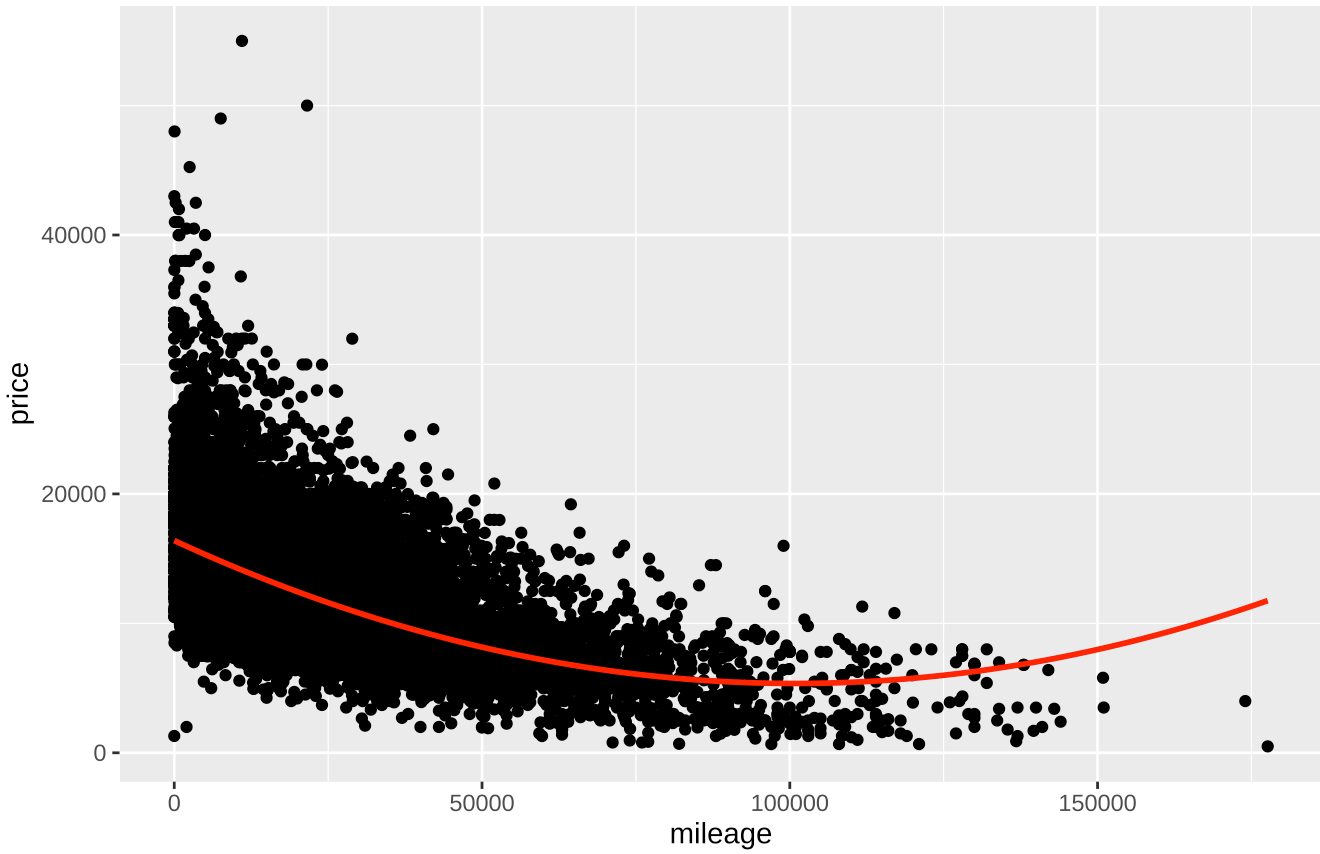
Comment: Using **Plot Bars** we are able to gather some insights from this data set. We see that **Ford Fiesta** and **Focus** are the models that have the higher frequencies compared to other models. Also, surprisingly **Manual Transmission** are overwhelmingly more frequent than automatic or semi-automatic cars. Finally, **Petrol** or also know as **Gasoline** is the most common fuel type, with **Diesel** coming in second.

When examining **Engine Size**, the majority of vehicles had smaller engines ranging from 1.0 to 2.0, which corresponds to the dataset's popularity of small cars such as the **Fiesta** and **Focus**. Furthermore, most cars have **low mileage**, which aligns with the histogram, which shows that the majority of vehicles were manufactured around 2020, representing their potential newer state. Finally, prices are predominantly focused in the lower range, with only a small proportion of automobiles falling into the upper-end price category. Moreover, we plotted for **correlation**, and we learned that our variables **Year**, **Mileage**, and **Engine Size** have a correlation with our target variable of the study, **Price**. First, **Year** with a correlation coefficient of **0.64**, has a positive correlation with **Price**, indicating that newer cars are generally more expensive, as expected. Also, **Engine Size**, has a positive relationship (**0.41**) with our target variable, indicating that bigger engines are associated with higher prices. Moreover, the relation between **Mileage** and **Price** is negative (**-0.53**), indicating that higher mileage in cars is expected to reduce its **Price**. Finally, we see that **Year** and **Mileage** also have an strong negative correlation (**-0.71**), indicating that newer cars tend to have lower mileage. This multicollinearity among predictors will be monitored during the modeling stage.

EDA: Consider 2 Variables that are non-linear

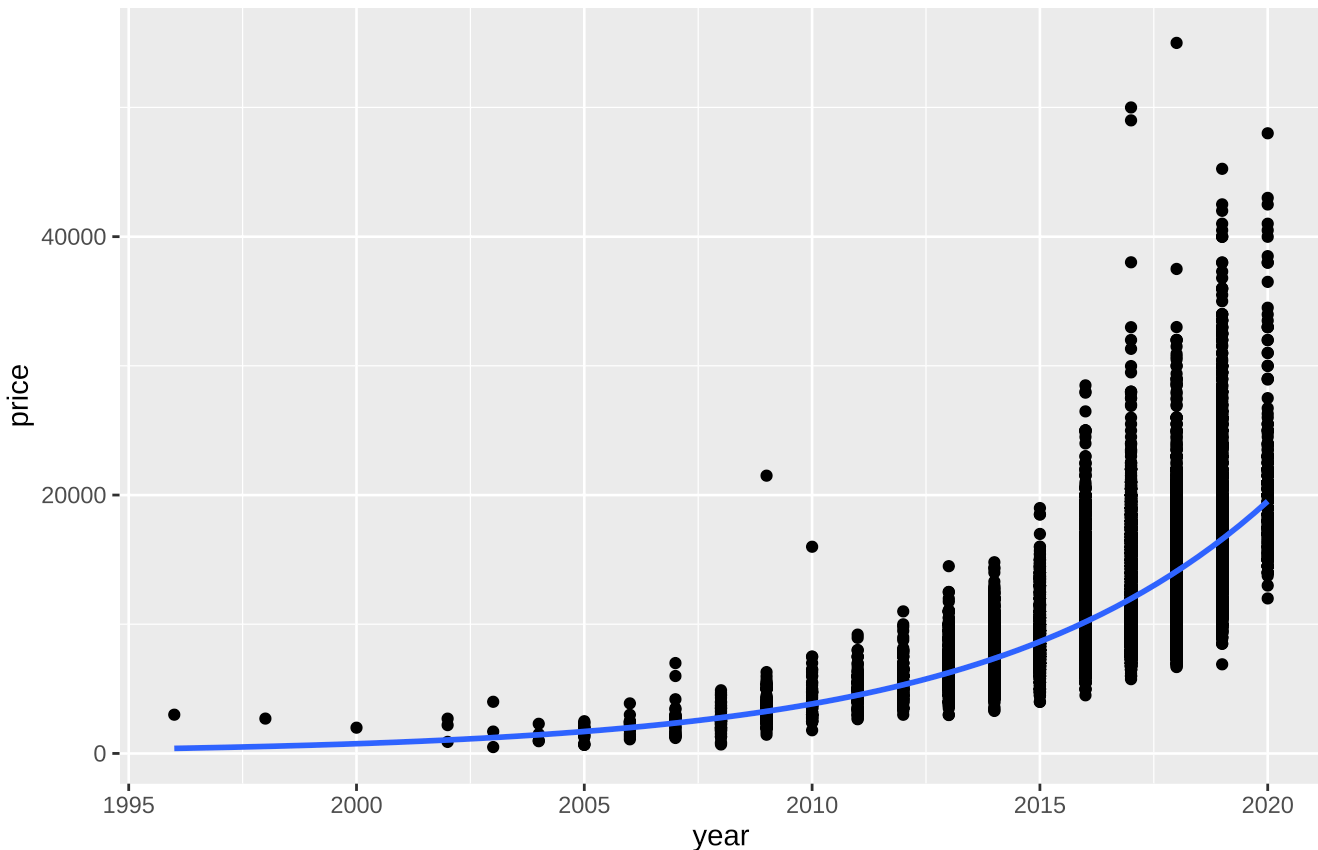
Relationship between Mileage and Price

With Quadratic Regression Lines



Relationship between adjusted Year and Price

Overlay Exponential Trendline



Comment: Plotting the **Relationship between Mileage and Price**, This confirms our earlier correlation analysis finding. It shows that as **mileage** increases, the vehicle's **price** generally decreases, indicating the projected depreciation pattern. Furthermore, the quadratic regression line shows how, as mileage keeps increasing and meets the higher mileage levels, the price tends to stabilize, implying that the marginal impacts of mileage on price decline as vehicles accumulate more miles.

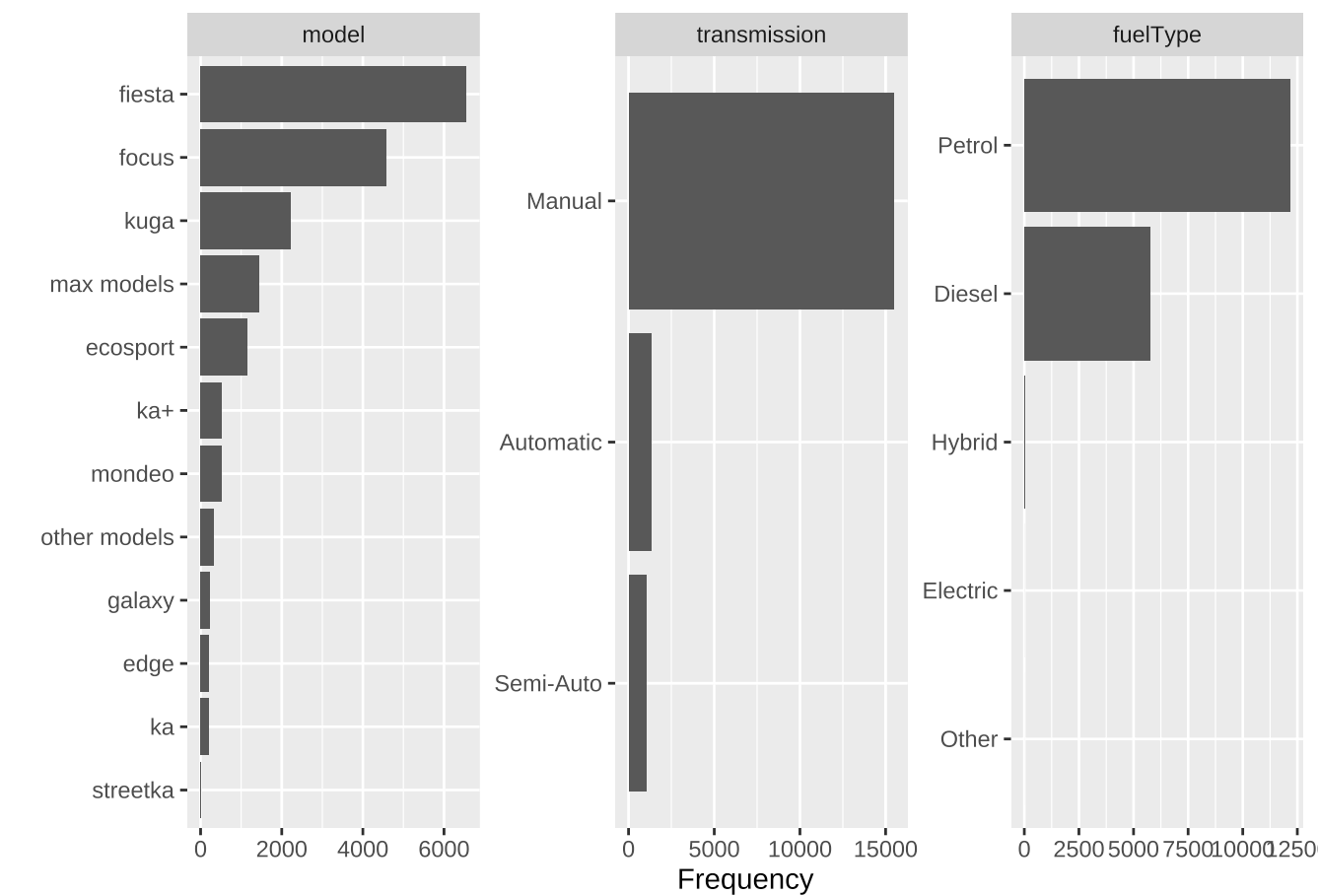
Moreover **Relationship between Year and Price** with an overlay exponential trend **upward**, reflects the higher values in prices are associated with newer models. This pattern reflects the higher **price** could be associated with newer models, as they are usually in better shape, incorporate more modern capabilities, and have **fewer mileage**.

DATA PREPARATION

Address outliers and missing values

	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize
1	fiesta	2017	12000	Automatic	15944	Petrol	150	57.7	1.0
2	focus	2018	14000	Manual	9083	Petrol	150	57.7	1.0
3	focus	2017	13000	Manual	12456	Petrol	150	57.7	1.0
4	fiesta	2019	17500	Manual	10460	Petrol	145	40.3	1.5
5	fiesta	2019	16500	Automatic	1482	Petrol	145	48.7	1.0
6	fiesta	2015	10500	Manual	35432	Petrol	145	47.9	1.6

	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize
17961	fiesta	2016	7999	Manual	31348	Petrol	125	54.3	1.2
17962	max models	2017	8999	Manual	16700	Petrol	150	47.1	1.4
17963	max models	2014	7499	Manual	40700	Petrol	30	57.7	1.0
17964	focus	2015	9999	Manual	7010	Diesel	20	67.3	1.6
17965	ka	2018	8299	Manual	5007	Petrol	145	57.7	1.2
17966	focus	2015	8299	Manual	5007	Petrol	22	57.7	1.0



Comment: In this stage, the only **data preparation** stage that is performed is the grouping of certain models for the benefit of our modeling stage. We grouped the **C-MAX**, **B-MAX**, **S-MAX**, and **Grand B-MAX**, under our **Max Models** group. The **Max** variations are typically more expensive than their regular counterparts, owing to variances in size, unique features, and increased seating capacity. Furthermore, models with little representation in the dataset have been grouped into a single category labeled **Other Models** to simplify the research and focus on more common vehicle types.

Partition the dataset

Partition the dataset 70% Training / 30% Validation and compare descriptives across all three

```
::: {.cell} ::: {.cell-output .cell-output-stdout}
```

```
[1] 12279.86
```

...

::: {.cell-output .cell-output-stdout}

[1] 12278.54

...

::: {.cell-output .cell-output-stdout}

[1] 12282.92

... :::

Comment: We confirm that the partitioning of the data is balanced and ready to move to the modeling stage.

MODEL DEVELOPMENT

Model 1: LIN-LIN REGRESSION MODEL WITHOUT INTERACTIONS

Call:

```
lm(formula = price ~ +year + mileage + mpg + tax + engineSize +  
  transmission + fuelType + model, data = trainSet)
```

Residuals:

Min	1Q	Median	3Q	Max
-12982.0	-1207.1	-125.6	1009.5	24132.8

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	-2373497.644239	26699.946172	-88.895
year	1183.503664	13.230858	89.450
mileage	-0.058530	0.001356	-43.153
mpg	-79.100496	3.145344	-25.148
tax	-0.592448	0.352178	-1.682
engineSize	3597.557246	72.489980	49.628
transmissionManual	-538.775525	70.456855	-7.647
transmissionSemi-Auto	-260.603716	95.382493	-2.732
fuelTypeElectric	1734.860939	1385.076373	1.253
fuelTypeHybrid	7584.396660	508.510193	14.915
fuelTypeOther	1127.570584	1954.647479	0.577
fuelTypePetrol	-58.461783	81.506890	-0.717
modelEdge	6697.020986	186.490836	35.911
modelFiesta	-425.524927	80.740416	-5.270
modelFocus	1370.002047	81.232642	16.865
modelGalaxy	4543.617917	178.705674	25.425

modelka	-2140.287223	183.083112	-11.690
modelka+	-4680.912299	124.070543	-37.728
modelkuga	1586.233910	96.283935	16.475
modelmax models	180.698748	97.293962	1.857
modelmondeo	678.123535	132.896753	5.103
modelother models	4427.920032	157.234819	28.161
modelstreetka	3418.503296	1955.914420	1.748

Pr(>|t|)

(Intercept)	< 0.0000000000000002	***
year	< 0.0000000000000002	***
mileage	< 0.0000000000000002	***
mpg	< 0.0000000000000002	***
tax	0.0925	.
engineSize	< 0.0000000000000002	***
transmissionManual	0.0000000000000221	***
transmissionSemi-Auto	0.0063	**
fuelTypeElectric	0.2104	
fuelTypeHybrid	< 0.0000000000000002	***
fuelTypeOther	0.5640	
fuelTypePetrol	0.4732	
modeledge	< 0.0000000000000002	***
modelfiesta	0.0000001384691392	***
modelfocus	< 0.0000000000000002	***
modelgalaxy	< 0.0000000000000002	***
modelka	< 0.0000000000000002	***
modelka+	< 0.0000000000000002	***
modelkuga	< 0.0000000000000002	***
modelmax models	0.0633	.
modelmondeo	0.0000003398504644	***
modelother models	< 0.0000000000000002	***
modelstreetka	0.0805	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1950 on 12555 degrees of freedom

Multiple R-squared: 0.8304, Adjusted R-squared: 0.8301

F-statistic: 2794 on 22 and 12555 DF, p-value: < 0.0000000000000002

GVIF Df GVIF^(1/(2*Df))

year	2.370617	1	1.539681
mileage	2.366412	1	1.538315
mpg	3.373876	1	1.836811
tax	1.584277	1	1.258681
engineSize	3.249866	1	1.802738
transmission	1.255777	2	1.058591
fuelType	5.166265	4	1.227855
model	3.957028	11	1.064518

Comment: With a **p-value** of <0.0000000000000002 and a **F-statistic** of 2.79e+03 we have a model that is statistically significant. Also, we are able to say that with an **R2** of 0.83, almost 83% of the variability of the data set is explained, which is very positive.

Moreover, we see that according to the model our variables Year, Mileage, and Engine Size are the most significant variables in determining vehicle prices, as their **p-value** appear to be extremely low. The **year of manufacture** is an important factor in deciding car prices, and it has a strong positive relationship with price, supporting the facts addressed throughout this research. As presented in the model results, for every additional yer (newer model), there is 1,183.50 dollar increase expected in price, meaning that the newer the model, the more expensive it will be.

Engine Size, also has a positive relationship with price, meaning that for every liter increase in engine size (liter capabilities of 1-5), the price is expected to increase by 3,597.56 dollars. This insight emphasizes the market values larger engines, which are frequently associated with more performance, power, and capability, higher than smaller engines.

On the other hand, **Mileage** has a negative relationship with price. For every additional mile driven, the car's price is predicted to drop by 0.06 dollars.his decrease reflects the natural depreciation associated with more miles, a shorter remaining lifespan, and perhaps higher maintenance costs, making the vehicle less appealing to potential buyers.

This model allows us to explore how other factors, such as **transmission type**, affect the pricing of a vehicle. Using **automatic transmission** as a baseline, it is clear that this is the **most expensive** choice. Cars with **semi-automatic** vehicles are 260.60. dollars less expensive than their automatic version, while vehicles with **manual transmissions** are the **cheapest**, priced \$538.77 less than automatic versions. This pricing trend is largely due to consumer preferences for the convenience and ease of use associated with automatic transmission.

In terms of **car models**, the **Edge** and **Galaxy** are the most expensive options. Using our **EcoSport** baseline model, the Edge model is estimated to cost 6,697.02 dollars more, while the Galaxy model will cost \$4,543.63. Furthermore, the **Focus** model costs 1,377.02 dollars more than the baseline. Finally, the "Other Models" category has a highly positive coefficient, as one would expect given that it includes high-value automobiles such as the Mustang, which is noted for its premium pricing.

Finally, when it comes to fuel type, the model shows that **hybrid vehicles** is the most expensive option, with an average price that is 7,584.39 dollars more than other fuel types. Similarly, **electric vehicles** cost 1,734.86 dollars more on average than other fuel types. In contrast, **petrol (gasoline)** cars are slightly less expensive, with an average price of \$58.46 less than other fuel types.

Multicollinearity

- We test for **Multicollinearity** and we see that based on **VIF** scores, all predictors are within acceptable limilts, which suggests that there is no issue here.

Model 2: LIN-LIN REGRESSION MODEL WITH INTERACTIONS

Call:

```
lm(formula = price ~ +year + mileage + mpg + tax + engineSize +  
  transmission + fuelType + model + year * mileage + mileage *  
  fuelType, data = trainSet)
```

Residuals:

Min	1Q	Median	3Q	Max
-13994.7	-1098.4	-67.8	955.3	23993.0

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value
(Intercept)	-3395993.763340	33807.977001	-100.449
year	1689.772025	16.745844	100.907
mileage	24.260158	0.558261	43.457
mpg	-53.921904	3.020561	-17.852
tax	-5.433712	0.343940	-15.798
engineSize	4139.092083	68.721707	60.230
transmissionManual	-704.767856	65.508598	-10.758
transmissionSemi-Auto	-159.912940	88.515237	-1.807
fuelTypeElectric	586.586292	3142.658804	0.187
fuelTypeHybrid	8110.003427	590.213887	13.741
fuelTypeOther	971.647307	1813.175917	0.536
fuelTypePetrol	121.422651	89.184016	1.361
modeledge	6946.479791	173.431432	40.053
modelfiesta	-681.126885	75.220296	-9.055
modelfocus	1052.540347	75.747296	13.895
modelgalaxy	4591.202329	165.793674	27.692
modelka	-2504.727743	170.226783	-14.714
modelka+	-4914.101605	115.396232	-42.585
modelkuga	1723.671666	89.530695	19.252
modelmax models	196.475846	90.268331	2.177
modelmondeo	447.421215	123.955031	3.610
modelother models	3990.353857	146.192463	27.295
modelstreetka	972.579239	1815.206081	0.536
year:mileage	-0.012070	0.000277	-43.570
mileage:fuelTypeElectric	0.040530	0.169045	0.240
mileage:fuelTypeHybrid	-0.083552	0.013316	-6.274
mileage:fuelTypeOther	NA	NA	NA
mileage:fuelTypePetrol	0.014084	0.001784	7.895

Pr(>|t|)

(Intercept)	< 0.0000000000000002 ***
year	< 0.0000000000000002 ***
mileage	< 0.0000000000000002 ***
mpg	< 0.0000000000000002 ***
tax	< 0.0000000000000002 ***
engineSize	< 0.0000000000000002 ***
transmissionManual	< 0.0000000000000002 ***
transmissionSemi-Auto	0.070846 .
fuelTypeElectric	0.851936
fuelTypeHybrid	< 0.0000000000000002 ***
fuelTypeOther	0.592050
fuelTypePetrol	0.173385
modeledge	< 0.0000000000000002 ***
modelfiesta	< 0.0000000000000002 ***
modelfocus	< 0.0000000000000002 ***

```

modelgalaxy          < 0.0000000000000002 ***
modelka              < 0.0000000000000002 ***
modelka+            < 0.0000000000000002 ***
modelkuga           < 0.0000000000000002 ***
modelmax models      0.029531 *
modelmondeo          0.000308 ***
modelother models    < 0.0000000000000002 ***
modelstreetka        0.592109
year:mileage          < 0.0000000000000002 ***
mileage:fuelTypeElectric 0.810522
mileage:fuelTypeHybrid 0.00000000036249319 ***
mileage:fuelTypeOther NA
mileage:fuelTypePetrol 0.00000000000000314 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 1809 on 12551 degrees of freedom

Multiple R-squared: 0.8541, Adjusted R-squared: 0.8538

F-statistic: 2826 on 26 and 12551 DF, p-value: < 0.00000000000000022

Comment: In this part, we improved the original model by include interactions between specific variables. This technique aims to study whether these interactions could improve the model's performance and provide deeper insights into the factors that influence vehicle prices. With a **p-value** of <0.0000000000000002 and a **F-statistic** of 2.83e+03 we have a model that is statistically significant. Also,we are able to say that with an **R2** of 0.854 ,almost 85.4% of the variability of the data set is explained, which is very positive, and slightly outperform our previous model.

Variable Significance: Variables in the model remain similar compared to the last model, based on significance and relationship.

Interactions:

By including interactions we are able to gather more insights:

- **Interaction between Year and Mileage:** This interaction is statistically significant, with a coefficient of -0.083552, indicating that this interaction has a negative relationship with **Price**. However, it shows that the negative effect that **Mileage** has on price, becomes slightly less pronounced as the year of manufacturing increases. This means that **newer cars** tend do a better job retaining some of the value, in comparison to other models.
- **Interaction between Mileage and Fuel Type:** This interaction helps us understand that with an **increase in Mileage, Hybrid** cars are the ones that have the higher negative effect on price. For, **Gasoline** cars, the effect is slightly less negative.

Therefore, incorporating interaction factors increased the model's fit and provided deeper insights into variable correlations. Also, the interactions demonstrated that newer automobiles hold their worth better despite more mileage, but hybrid autos face a stronger negative pricing impact as mileage increases. This helps us have a more detailed understanding of the factors that influence vehicle costs.

Model 3: LIN-LIN REGRESSION MODEL WITH INTERACTIONS AND YEAR AS QUADRATIC

Call:

```
lm(formula = price ~ +poly(year, 2, raw = TRUE) + mileage + mpg +
    tax + engineSize + transmission + fuelType + model + engineSize *
    mileage + mileage * fuelType, data = trainSet)
```

Residuals:

Min	1Q	Median	3Q	Max
-17840.1	-1086.3	-78.7	991.8	24829.3

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value
(Intercept)	355078650.070383	7876397.345660	45.081
poly(year, 2, raw = TRUE)1	-353705.371593	7819.727739	-45.232
poly(year, 2, raw = TRUE)2	88.085206	1.940855	45.385
mileage	-0.005032	0.005280	-0.953
mpg	-58.629669	2.984210	-19.647
tax	-4.907302	0.372207	-13.184
engineSize	4616.986229	80.116319	57.629
transmissionManual	-691.267093	65.079737	-10.622
transmissionSemi-Auto	-157.096338	87.871205	-1.788
fuelTypeElectric	1002.421838	3119.115253	0.321
fuelTypeHybrid	8293.226285	585.210612	14.171
fuelTypeOther	1134.491778	1799.601251	0.630
fuelTypePetrol	503.970636	91.741715	5.493
modeledge	6919.709502	172.091811	40.209
modelfiesta	-697.335835	74.759005	-9.328
modelfocus	1040.017932	75.293646	13.813
modelgalaxy	4680.158733	164.814212	28.397
modelka	-2571.754644	169.034163	-15.214
modelka+	-4849.356011	114.480607	-42.360
modelkuga	1704.173789	88.884198	19.173
modelmax models	254.303443	89.696842	2.835
modelmondeo	532.690129	124.519780	4.278
modelother models	3598.398123	145.885034	24.666
modelstreetka	-1291.570325	1803.720869	-0.716
mileage:engineSize	-0.029222	0.002871	-10.178
mileage:fuelTypeElectric	0.032091	0.167780	0.191
mileage:fuelTypeHybrid	-0.087166	0.013220	-6.593
mileage:fuelTypeOther	NA	NA	NA
mileage:fuelTypePetrol	-0.005597	0.002327	-2.405

Pr(>|t|)

(Intercept)	< 0.0000000000000002 ***
poly(year, 2, raw = TRUE)1	< 0.0000000000000002 ***
poly(year, 2, raw = TRUE)2	< 0.0000000000000002 ***
mileage	0.34063
mpg	< 0.0000000000000002 ***
tax	< 0.0000000000000002 ***

```

engineSize          < 0.0000000000000002 ***
transmissionManual  < 0.0000000000000002 ***
transmissionSemi-Auto      0.07383 .
fuelTypeElectric      0.74793
fuelTypeHybrid        < 0.0000000000000002 ***
fuelTypeOther         0.52844
fuelTypePetrol        0.0000000402032 ***
modeledge            < 0.0000000000000002 ***
modelfiesta          < 0.0000000000000002 ***
modelfocus           < 0.0000000000000002 ***
modelgalaxy          < 0.0000000000000002 ***
modelka              < 0.0000000000000002 ***
modelka+             < 0.0000000000000002 ***
modelkuga            < 0.0000000000000002 ***
modelmax models      0.00459 **
modelmondeo          0.0000190014529 ***
modelother models    < 0.0000000000000002 ***
modelstreetka        0.47397
mileage:engineSize    < 0.0000000000000002 ***
mileage:fuelTypeElectric  0.84832
mileage:fuelTypeHybrid  0.000000000000447 ***
mileage:fuelTypeOther      NA
mileage:fuelTypePetrol  0.01617 *
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1796 on 12550 degrees of freedom

Multiple R-squared: 0.8563, Adjusted R-squared: 0.856

F-statistic: 2769 on 27 and 12550 DF, p-value: < 0.00000000000000022

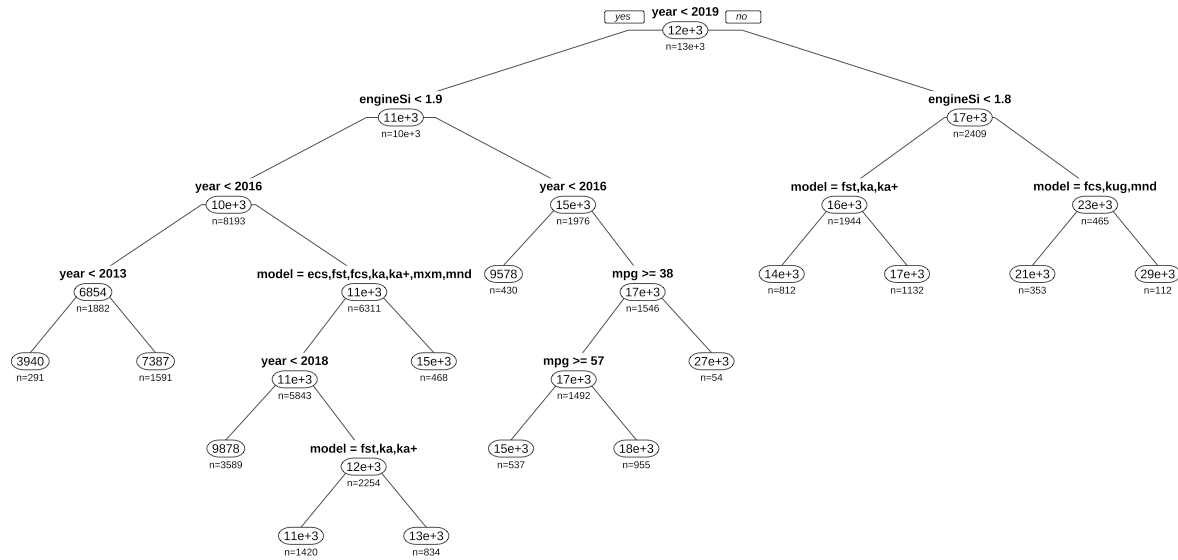
Comment: Now in this part, we improved the previous model by including our variable **Year** as **Quadratic**, while also including interactions between specific variables. This technique aims to study whether these interactions could improve the model's performance and provide deeper insights into the factors that influence vehicle prices. With a **p-value** of <0.0000000000000002 and a **F-statistic** of 2.83e+03 we have a model that is statistically significant. Also, we are able to say that with an **R2** of 0.856, almost 85.6% of the variability of the data set is explained, which is very positive, and slightly outperform our previous model, however the change is not big.

Difference - The addition of quadratic term for year creates a non-linear relationship with price, allowing for a deeper comprehension of how newer models effect pricing. The primary effect of mileage is no longer significant but its interactions with other variables, particularly engine size and fuel type, are still crucial. While most variables maintain their former importance and interactions, this updated model provides a better fit.

Model 4: REGRESSION TREE

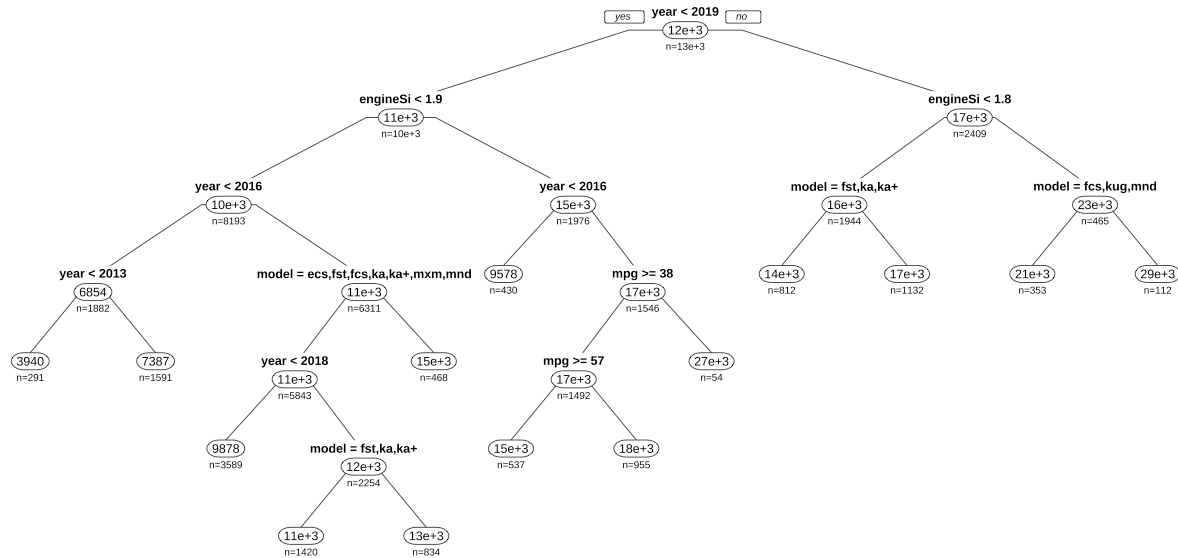
Generate the Default Tree

DEFAULT TREE



Find the Optimal Tree

FULL TREE



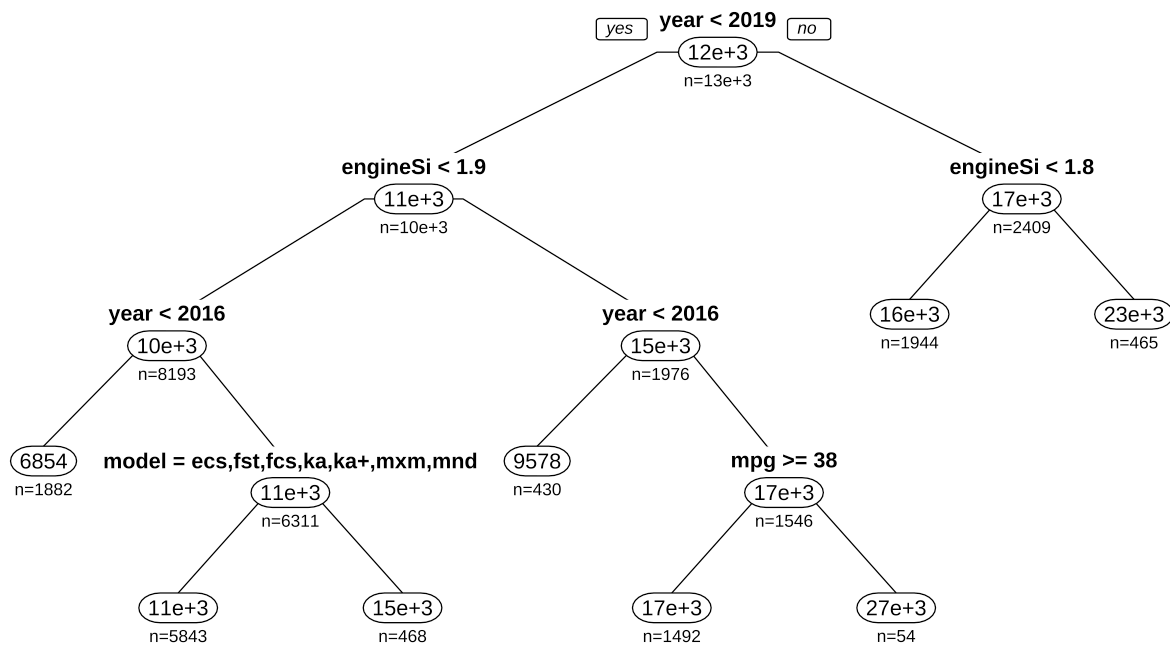
FULL TREE – Identify the complexity parameter (cp) associated with the smallest cross-validated prediction error

	CP	nsplit	rel error	xerror	xstd
1	0.27797553	0	1.0000000	1.0002292	0.019804776
2	0.15553413	1	0.7220245	0.7222810	0.015772740
3	0.08837796	2	0.5664903	0.5669538	0.013484262
4	0.06601537	3	0.4781124	0.4785678	0.013153853
5	0.06369376	4	0.4120970	0.4195412	0.011237350
6	0.03321688	5	0.3484032	0.3488170	0.009329122
7	0.02010003	6	0.3151864	0.3156394	0.009111969
8	0.01984693	7	0.2950863	0.2974914	0.008762026
9	0.01943784	8	0.2752394	0.2883358	0.008151858
10	0.01849974	9	0.2558016	0.2655825	0.007769688

Comment: Based on the cross-validation prediction error and the model's simplicity, tree **number 8** is the ideal tree. This tree has **7 splits** with a cross-validation error (xerror) of **0.2974914**, indicating a model with a good combination of performance and complexity.

Create the Optimal Tree

OPTIMAL TREE



Comment: Based on our **Optimal Tree**, again, **year** is a crucial variable when determining **car price** as the first and most important split occurs with the year of manufacture, which separates cars made before 2019 from newer models. For vehicles made before to 2019, they meet another split that establish **Engine Size** as a relevant predictor. Cars with smaller engine capacity (less than 1.9 liters) are a separate category with lower average pricing than those with larger engines, which strengthens our position through the study, as smaller engines are expected to be related with lower prices.

For this cars with smaller engines, additional division based on year < 2016 further distinguishes older models and cheaper pricing. Among these older vehicles, EcoSport (ecs), Fiesta (fst), Focus (fcs), Ka, Ka+, and Mondeo (mnd) have moderate pricing discrepancies, reflecting variations in model-specific demand and characteristics.

Moreover, cars with engine capacity less than 1.8 liters are classified separately, with lower average prices than those with bigger engines. Also, **Miles per gallon (mpg)** appears for the first time as a crucial predictor, since vehicles with higher fuel efficiency ($\text{mpg} \geq 38$) are separated further, reflecting a premium for fuel-efficient cars in this range. Notably, a very small subset of high-mpg automobiles commands much higher prices.

MODEL EVALUATION

Evaluation and comparison of all 4 models based on RMSE and MAPE

Model 1: Full Linear – RMSE & MAPE

[1] \$1,999

[1] 14.66%

Model 2: With Interaction Effects – RMSE & MAPE

[1] \$1,871

[1] 12.96%

Model 3: Interaction Effects and Quadratic – RMSE & MAPE

[1] \$1,855

[1] 12.69%

Model 4: Regression Tree – RMSE & MAPE

[1] \$2,648

[1] 19.00%

Comment: Among the 4 models, **Number 3**, which is the Regression model with Interaction Effects and Quadratic Terms, demonstrates the best overall performance. With an **RMSE** of \$1,855, it establishes itself as the best performer in this metric. Also, it has the lowest **MAPE** with 12.69%, showing its accuracy in predicting vehicle prices. Also, it is important to remember that this model is **statistically significant**, and has an **adjusted R2** of **0.856**, which shows that the model does a great job in explaining the variability of the data set.

DEPLOYMENT

For the deployment and reproducibility of this model, I recommend utilizing the **Linear Regression Model with Interaction Effects and Quadratic Year**, which provided the best performance metrics among the four models. Its better explanatory power and capacity to capture hidden interactions between variables make it the best fit for deployment. Moving forward, this model may be used to provide precise and data-driven insights on vehicle price, and it can be updated as new data becomes available. Furthermore, including this model within a dynamic pricing system or decision-support tool may increase its usefulness and adaptability to changing market conditions.

RECOMMENDATIONS

Dear Mr. Felich,

I hope this message finds you well.

Through the development and analysis of the prediction models, we were able to identify the key factors influencing the price of Ford vehicles, providing valuable insights for pricing strategies and market positioning. Among these, the **year of manufacture** is most important, with newer models commanding greater prices. The price rises by around 1,183.50 per year, indicating consumer demand for current features, improved safety, and advanced technology, also results in a respective increase in price. Similarly, **Engine size** also being highly significant, has a positive effect on pricing, with each one-liter increase resulting in an average price increase of \$3,597.56. This finding highlights the premium buyers prefer might prefer larger engines, most likely due to their perceived performance and power.

Based on model-specific research finds that the Edge and Galaxy versions are among the most expensive, with prices 6,697.02 and 4,543.63 higher, respectively, than the standard EcoSport. The Focus model also charges a premium, costing \$1,377.02 more than the standard. However, Fiesta, our most recurrent model is 681 dollars cheaper than our baseline car.

On the other hand, **mileage** has a negative impact on price, with each additional mile driven reducing the vehicle's value by 0.06. This aligns with the expected depreciation of used vehicles over time. However, the interaction between year and mileage highlights that newer cars retain their value better, even at higher mileage levels, however, when the car starts getting older, its depreciation starts incrementally increasing.

Furthermore, interaction effects reveal that **hybrid vehicles** are more sensitive to mileage increases than other fuel types, as are the ones that seem to get harmed the most with mileage increase.

Gasoline cars appears to have the lowest decrease in price, with mileage incremental.

Therefore, based on this findings and the current stock we have a high stock number, you should try to effectively reach college students and young adults with limited budgets, I recommend establishing a targeted marketing campaign that emphasizes the Fiesta and Focus models' cost, practicality, and lifestyle appeal. Using the names of these models, the campaign can include innovative wording such as "Drive the Fun" for the Fiesta, appealing to youthful energy and social lifestyles, and "Stay Focused on Your Goals" for the Focus, emphasizing ambition. I also recommend to prioritize digital channels such as Instagram, TikTok, and YouTube and Collaborations with micro-influencers who are students or young professionals can help to increase authenticity and relatability. Furthermore, on-campus activities and partnerships with universities can foster direct involvement by providing test drives and exclusive student discounts.

Best regards, Jose Almanza

CITATION

R Version and Packages

R version 4.3.1 (2023-06-16)

R Packages Used:

```
[1] "dplyr"      "readxl"      "summarytools" "corrplot"     "dlookr"  
[6] "knitr"      "formattable" "DataExplorer" "forecast"     "rpart.plot"  
[11] "rpart"      "caret"       "lattice"      "ggplot2"
```

OTHER REFERENCES

Quku, A. (2022, April 11). Ford car price prediction. Kaggle.

<https://www.kaggle.com/datasets/adhurimquku/ford-car-price-prediction>